

The Decoupling Locus: a MOND Scale Derived from a Clock, and the Solar System Reduced to One Number

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Abstract

A cuscuton-clock sector with a projected-gradient scalar produces a cold clustering component and passes nine cosmological and solar-system gates as a dark component [3, 4, 5, 6], but produces no force law, because matter in that action is minimally coupled: $\delta S_{\text{matter}}/\delta\chi = 0$ identically. Adding a coupling is what would give MOND, and the obstruction was that the clock and the scalar mix, so a coupled matter source would drag the clock and generate preferred-frame effects. We report that the mixing vanishes *exactly* on the locus $W_0 \equiv U - 2\gamma\bar{q}^2\dot{\bar{q}} = 0$, which is a condition on coefficients the action already has. A first pass priced that locus at a cubic coupling eight orders too large, scaling as $1/w$; that pricing held the clock coefficient U fixed, and U is not free. With $\rho = U/m_{\text{rel}}$ imposed the factor of w cancels identically: what the locus fixes is the dimensionless ratio $\gamma\bar{q}^3 H/\rho = -1/[6(s_0 - 1)]$, of magnitude at most $1/6$, and the operator's weight against gravity is $\Omega_{\text{sec}}/[6(s_0 - 1)] \leq 0.044$. The earlier verdict is withdrawn. The locus also *forces* two previously free quantities: positivity of the sector's remaining energy requires $s_0 \geq 2$, hence $m_{\text{rel}} \leq w \lesssim 10^{-4}$. Extending the action by the $Y^{3/2}$ operator identified as missing in [7] and a leading-order matter coupling, the quasi-static gradient sector yields the deep-MOND scaling outright ($d \log g/d \log r = -1$, $d \log g/d \log M = +1/2$) and fixes the acceleration scale, $a_0 = \lambda^3/(12\pi G\beta s_0)$, mass-independent and falling with the clock rate. The single conflict — the branch wants the linear gradient coefficient large, deep MOND wants it small, and it is the same coefficient — resolves into one inequality, $C \equiv 2k\bar{q}^2/U \gtrsim 10^3$, whose derived exponents cancel exactly, so C is a constant of the motion. Finally, coupling matter *disformally* along the clock's own unit timelike direction shifts both weak-field potentials equally, giving $\gamma_{\text{PPN}} = 1$ exactly with no added vector field; boosted, the same term gives $\alpha_1 = 8f_s$, five orders over the bound unless local matter drags the clock to within 4.6 m s^{-1} . That alignment, and κ , are what remain open. Ten new Lean 4 certificates; the file stands at 124 theorems, zero **sorry**. *Version 2 (18 September 2026) adds an erratum: the disformal matter coupling of §5 is excluded by GW170817's photon-graviton arrival test by $3\text{--}4 \times 10^7$, §4–§5 are withdrawn, §2–§3 stand, and the single-metric replacement is named.*

1 Position, and three corrections to the previous paper

The action [2] is, with $c = 1$ and signature $(-+++)$,

$$S = \int d^4x \sqrt{-g} \left[\frac{M^2}{2} (R - 2\Lambda) + P(X, \tau) - V(\tau) + sW(Y, \tau) + \gamma X \square \chi \right] + S_r + S_b, \quad (1)$$

with $s = \sqrt{-\nabla_\mu \tau \nabla^\mu \tau}$, $n_\mu = -\partial_\mu \tau / s$, $Q = n^\mu \partial_\mu \chi$, $Y = (g^{\mu\nu} + n^\mu n^\nu) \partial_\mu \chi \partial_\nu \chi$, $X = Q^2 - Y$. An external audit of [7] found three errors, each reproduced independently before being accepted, and they are recorded here because two of them were load-bearing.

(i) *The health condition had the wrong sign.* We stated $4dl > U$. Checking the same reduction against the canonical scalar $P(X) = X/2$, which gives $F_Y = -\frac{1}{2}$, shows a healthy field has *negative* F_Y . The correct condition is $U > 4dl$. Every verdict resting on the reversed inequality is withdrawn, including a proposed coefficient window.

(ii) *The interpolating exponent was measured against the wrong acceleration.* The reported slope 0.470 compared the kernel with the Newtonian acceleration $x = g_N/a_0$ rather than the true one $y = xv(x)$. Converted, the slope is 0.997. The qualitative conclusion of [7] — that MOND requires a linear μ in y , which the unextended action does not supply — survives with the number corrected.

(iii) *The no-force-law verdict was right for a different reason.* [7] excluded a force law on the exponents of the two gradient operators. The stronger and simpler fact is that matter in (1) enters only through S_r and S_b , both minimally coupled to $g_{\mu\nu}$: $\delta S_{\text{matter}}/\delta \chi = 0$ identically. The action modifies *no* force law, and the rotation-curve exponents of [7] are conditional on a coupling that is not present. Correspondingly, a structural argument that a vanishing background gradient removes the clock–scalar mixing is also withdrawn: the exact quadratic action carries a cross term $-2qW_Y$ and a clock response $\sigma/\pi = 2qsW_Y/(2q^2W_Y - W_0)$, both of which survive $Y \rightarrow 0$. The preferred-frame gate returns to open, and that is the gate this paper addresses.

2 The locus on which the clock stops listening

The clock response above vanishes identically, together with the entire W_Y contribution to the eliminated scalar coefficient, when

$$W_0 \equiv U - 2\gamma\bar{q}^2\dot{\bar{q}} = 0, \quad (2)$$

at which point that coefficient reduces to $2P_X$ alone. Since W_0 is built from quantities (1) already contains, this is a locus inside the existing parameter space, not a modification. It has one immediate cost: with $W_0 = 0$ every surviving term of the clock equation carries the same factor of the clock rate, which therefore cancels. That equation no longer determines s_0 but constrains the coefficient functions; s_0 remains fixed by conservation through $s_0 - 1 = w/m_{\text{rel}}$.

3 What the locus costs, measured in units that cannot be inflated

The stress tensor, derived rather than quoted. Write $S = \int dt Na^3 L_m$ in minisuperspace with the lapse restored. Then $\rho = -a^{-3} \partial_N (Na^3 L_m)|_{N=1}$ and $p = (3a^2)^{-1} [\partial_a - d_t \partial_{\dot{a}}] (Na^3 L_m)|_{N=1}$. Calibrating on $L_m = F(Q)$ reproduces the textbook pair $QF_Q - F$ and F . Applied to the cubic operator, whose minisuperspace action is $-2\gamma a^2 \dot{a} Q^3$ after an integration by parts that differs from the unintegrated form by an explicit total derivative,

$$\rho_3 = -6\gamma H \bar{q}^3, \quad p_3 = 2\gamma \bar{q}^2 \dot{\bar{q}}, \quad (3)$$

so that $W_0 = U - p_3$: *the decoupling condition says the cubic operator's pressure equals the clock coefficient.* On the derived family $\bar{q} \propto a^{-3w}$ one finds $p_3 = w\rho_3$ identically, so the operator carries the sector's own equation of state and switching it on does not disturb the background scaling.

The correction. Using the family's $\rho = U/m_{\text{rel}}$ and $p = U(s_0 - 1)$ [5], the cubic operator's share of the sector density is

$$\frac{\rho_3}{\rho} = \frac{1}{s_0 - 1}, \quad (4)$$

free of both U and w . A first pass had reported $|\gamma| = U/(6|w|\bar{q}^3 H)$ and concluded, holding U fixed, that the acoustic bound $w \lesssim 10^{-4}$ forces $|\gamma| \sim 650$ against the 10^{-6} used throughout. But $U = m_{\text{rel}}\rho$ with $m_{\text{rel}} = w/(s_0 - 1)$, and substituting, $\partial|\gamma|/\partial w = 0$ exactly. The divergence was an artifact of units in which the sector density had been set to $1/m_{\text{rel}}$, which is large for the same reason w is small. What the locus fixes is a ratio,

$$\frac{\gamma\bar{q}^3 H}{\rho} = -\frac{1}{6(s_0 - 1)}, \quad (5)$$

of magnitude at most $1/6$; today $|\gamma|\bar{q}^3 = 1.13 \times 10^{21} \text{ eV}^3$, a cube root of 10.4 MeV . Comparing the cubic operator's share of the momentum conjugate to the scale factor with the Einstein–Hilbert term's, from the same action,

$$\left| \frac{\pi_3}{\pi_{\text{grav}}} \right| = \frac{\Omega_{\text{sec}}}{6(s_0 - 1)} \leq 0.044, \quad (6)$$

a percent-level correction. Every result derived at $\gamma \rightarrow 0$ therefore survives on this locus to about four percent, and the earlier verdict that they were invalidated is withdrawn.

Two free quantities become forced. The non-cubic density is $\rho[1 - 1/(s_0 - 1)]$, which is non-negative only for

$$s_0 \geq 2 \quad \Longleftrightarrow \quad m_{\text{rel}} \leq w \lesssim 10^{-4}. \quad (7)$$

The clock must run at least twice proper time, and the margin must sit *below* the equation of state rather than above it. At the boundary $s_0 = 2$ the cubic operator supplies the entire dark sector density on its own. Both statements are Lean-certified.

4 The force law on the locus

Extend W by the $Y^{3/2}$ operator [7] identified as the missing ingredient, and couple matter at leading order, $\lambda\varphi\rho_m$. In the quasi-static limit the gradient equation is $\nabla \cdot (2sW_Y \nabla\varphi) = -\lambda\rho$, so $2sW_Y$ plays the role of the interpolating function. With $W = \beta Y^{3/2}$ the spherical solution gives an acceleration on matter with

$$\frac{d \log g}{d \log r} = -1, \quad \frac{d \log g}{d \log M} = +\frac{1}{2}, \quad (8)$$

the deep-MOND scaling, obtained from the action rather than assumed. Matching to $\sqrt{GMa_0}/r$ fixes the scale in terms of the action's own coefficients,

$$\boxed{a_0 = \frac{\lambda^3}{12\pi G\beta s_0}} \quad \text{with} \quad \frac{\partial a_0}{\partial M} = 0, \quad (9)$$

so it is a constant of the theory rather than an object-by-object fit, and it *falls as the clock runs faster*. On the canonical footing $\lambda^3/(\beta s_0) = 2.3555 \times 10^{-19}$ in SI units and on the alternative footing 2.8378×10^{-19} , the ratio being the footing fork 1.204777 itself.

The one conflict, and its window. The linear term d in W_Y appears twice: as the floor $\mu_{\text{floor}} = d/k$ it places under the interpolating function, k being W_Y at the highest acceleration probed, and as the closure's $\mu = 2d\bar{q}^2/U$ that sets the clock margin. They are the same d , related by $\mu = C \mu_{\text{floor}}$ with $C = 2k\bar{q}^2/U$. The locus needs $\mu \geq 1 - w$; the rotation-curve data need $\mu_{\text{floor}} \leq 10^{-3}$ for the radial acceleration relation to survive at the ten-percent level down to $g/a_0 = 10^{-2}$. Both hold exactly when

$$C = \frac{2k\bar{q}^2}{U} \gtrsim 10^3. \quad (10)$$

On the derived family the exponents cancel, $-3(1 - w) - 6w + 3(1 + w) = 0$, so C is a constant of the motion: imposed once, it holds for all time. Missing the window does not kill the construction but buys it with a cancellation among the sector's densities of degree m_{rel}/w , which reaches 10^4 at $C = 1$ — the same $1/w$ the first pass saw, relocated from the coupling to the cancellation, where it belongs.

5 The solar system, reduced to one number

A *conformal* matter coupling shifts the time potential and not the space one and fails light bending outright. The standard cure is a *disformal* coupling along a unit timelike vector, which TeVeS [10] and AeST [11] must add as a separate dynamical field. Action (1) already has one: the clock's gradient supplies n_μ , and the cusciton [9] propagates no scalar of its own. Expanding $\tilde{g} = (1 - 2\varphi)g - 4\varphi n \otimes n$ in the weak field,

$$\tilde{\Phi} = \Phi + \varphi, \quad \tilde{\Psi} = \Psi + \varphi \quad \implies \quad \gamma_{\text{PPN}} = 1 \text{ exactly}, \quad (11)$$

however strong the scalar: the shift is common to both potentials and cancels from their difference. Light bending and Shapiro delay are general-relativistic.

Boosting is where it costs. For a system moving at w through the clock's frame the same term gives $\tilde{g}_{0i} = 4\varphi w_i$, first order in the velocity *and* first order in the scalar, which is the preferred-frame signature. Matching to the post-Newtonian form,

$$\alpha_1 = 8f_s, \quad (12)$$

f_s being the scalar's share of the local potential. At the share MOND requires at solar-system accelerations, $f_s \approx 1$, that is 8 against a bound of 10^{-4} : over by 8×10^4 . A clock sitting in the cosmic frame is excluded outright, by the same wall that closed the vector-based completions. The bound converts directly into an alignment requirement:

$$\boxed{v_{\text{residual}} \leq 4.6 \text{ m s}^{-1}} \quad (13)$$

between the clock's frame and the local matter frame, one part in 8×10^4 of the solar system's 370 km s^{-1} through the cosmic frame. *Whether the clock's own field equation delivers that is not computed here*, and an earlier lane measuring a cusciton clock as dragged at $3 \times 10^{-2} \Phi_N$ at Saturn [2] is the same question in a different parameterisation and was recorded there as a kill.

6 The board, and what is not claimed

Every gate the programme has posed now carries a computed number on this locus — the preferred-frame mixing (identically zero), the cubic backreaction (≤ 0.044), the clock rate (≥ 2) and margin

($\leq w$), the acoustic scale, the forest through criticality [4], lensing and S_8 (the sector is inert), and the force law and its scale — with two exceptions, which are stated as exceptions. The clock alignment above is not computed. And $\kappa = \frac{1}{2}$ remains fitted and provably underivable by this class of actions [1]: that is a theorem of this programme, not a gap in this paper. **This is not a complete theory of gravity and nothing here should be read as one.**

Limits, in full. The elimination giving the mixing coefficient is an external audit's, used rather than re-derived. The positivity argument forcing $s_0 \geq 2$ assumes no piece of the sector carries negative energy; if one may, $s_0 < 2$ costs a cancellation of degree $(2 - s_0)/(s_0 - 1)$. The weight π_3/π_{grav} is a background measure; the perturbation kinetic matrix at $s_0 = 2$ has not been recomputed and no gate has been re-run numerically on the locus. The matter coupling is leading order, not a full conformal or disformal factor; the quasi-static reduction drops the clock's gradient response, legitimate only because the mixing vanishes here; k is defined, not derived; the interpolation used is the pure $Y^{3/2}$ limit, not ν_{RAR} in full, so the deep-MOND limit and the scale are established and the whole interpolating function is not. The post-Newtonian matching uses the standard normalisation, so the coefficient 8 is convention-dependent at the factor-of-two level; the five-orders conclusion is not.

Erratum (version 2, 18 September 2026): the photon–graviton gate, and what replaces §5

The omission. Section 5 couples matter to $\tilde{g}_{\mu\nu} = g_{\mu\nu} + B n_\mu n_\nu$ along the clock's unit timelike direction so that light bends by $\tilde{\Phi} = \Phi + \varphi$. Photons then propagate on $\tilde{g}$ while gravitational waves propagate on g , and GW170817/GRB170817A bounds the accumulated photon–graviton arrival difference along the 40 Mpc sightline to the observed $+1.74 \pm 0.05$ s. That gate was not run here, although the repository already contained it (`prep_2026/gw170817_check`, which excluded the same two-cone structure at 3.5×10^7 s for an earlier deposit). Run for this construction (`fable_independent_2026/clock_swarm_2026/CK01_clock_gw170817_gate.py`, 5/5), with the scalar potential $\varphi' = [f_s + \nu(x) - 1] g_N/c^2$, the Milky Way and NGC 4993 as the two mandatory crossings and the external-field cap $\eta_{\text{ext}} = 0.03$, the delay is $5.8\text{--}7.4 \times 10^7$ s, i.e. 1.8–2.3 yr, on both a_0 footings and both kernels. The deep-MOND part of φ alone ($f_s = 0$) already gives 1.8 yr, so no choice of the Newtonian-regime share helps; the sign is the observed one (photons later); a 10 s emission budget would require a total scalar share below 2×10^{-7} of the potential, i.e. no MOND force from the scalar at all. **Section 5 is withdrawn**, and with it the force-law normalisation of §4, $a_0 = \lambda^3/(12\pi G\beta s_0)$, which belongs to the same coupling, and the flat- a_0 drift ceiling on w derived from it in the accompanying lane L217 (the acoustic ceiling $w \lesssim 10^{-4}$ stands). Sections 2–3 — the locus $W_0 = 0$, $\rho_3/\rho = 1/(s_0 - 1)$, $s_0 \geq 2$, $|\pi_3/\pi_{\text{grav}}| \leq 0.044$ — are statements about the sector's background and stand, as do the cosmological results of the preceding papers.

The split cannot be moved. For $\tilde{g} = g + B n n$ the photon and graviton cones differ by $B/2$ whichever sector carries B : moving the disformal term into the gravitational action only reverses the sign of the split (`CK02_replacement_coupling.py`, 6/6). The only cure is one metric for every species.

What replaces §5 (constructive). With matter, photons and gravitons on one metric, the force law must be sourced through the clock's own constraint (the mechanism of AeST carried by the clock, matter minimally coupled to g), with the scalar's Newtonian-regime $1/r$ field screened inside a healing length ξ by a $\xi^2(\nabla V)^2$ operator. That construction exists on the record (`closure_2026/`

FINAL_THEORY_CANDIDATE_2026-09-05.md; lanes f30–f35 and g03b–g03h) and passes by computation: the Cassini quadrupole (static floors 0.03/0.05 pc; ephemeris floor $\xi \geq 4.00$ pc), $\gamma_{\text{PPN}} = 1$, $\alpha_1 = -4c_{14}$ with the scalar’s drag $-4(2 - K_B)/[J_Y(1 + \xi^2 k^2) + 1]$ below 10^{-9} at Saturn for $\xi = 4$ pc, linear health at the corner, $c_T = c$, and, being single-metric, the GW170817 gate identically. It needs no clock alignment, which is why the companion paper’s requirement $s_0 \gtrsim 1.5 \times 10^7$ is withdrawn there. Its own open gates — the Ward identity, the Dirac count, FLRW with the clock, one number for $\xi_{\min}$, the α_2 reconciliation, causality, cosmological perturbations, the galactic solve — are listed with contracts in `clock_swarm_2026/CLOCK_WORK_ORDER.md`. Nothing in this erratum derives κ .

7 Reproducibility

fable_independent_2026/ lanes L211_three_corrections.py, L212_decoupling_branch.py (commit 473860435), L213_branch_solved_and_the_board.py, L214_force_law_on_the_branch.py (commit 425a28355) and L215_solar_system_on_the_branch.py (commit bde5bfd2c), each with its committed output and results file. The Lean file fable_independent_2026/lean_2026/Mondlean.lean carries cubic_share_is_clock_rate, positivity_forces_clock_rate, margin_below_equation_of_state, cubic_coupling_is_equation_of_state_independent, mond_scale_mass_independent, branch_mond_window, derived_exponents_cancel, window_ratio_is_scale_invariant, disformal_gamma_is_one and preferred_frame_alignment_requirement; 124 theorems, exit 0, zero sorry, axioms $\subseteq \{\text{proptext}, \text{Classical.choice}, \text{Quot.sound}\}$.

References

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